# 1. Clinical Question & Core Hypothesis

## Clinical Question

The Ontologist highlights a tension where DNA-repair deficiency (MSH6 loss) and high mutation load (TMB) should increase tumor immunogenicity, yet many ostensibly immune-hot bladder tumors fail to benefit from immunotherapy. Why does high neoantigen potential not consistently translate into effective antitumor immunity?

## Working Hypothesis

In bladder cancer, TLS-rich tumors that harbor MSH6 deficiency and high tumor mutational burden exhibit heterogeneous responses to immune checkpoint blockade because tumor-intrinsic eIF4E activity reprograms the translational landscape to favor immunosuppressive programs. Specifically, elevated eIF4E expression leads to preferential translation of immunomodulatory mRNAs (e.g., PD-L1, cytokines, metabolic enzymes) that dampen TLS-driven anti-tumor immunity, resulting in resistance to PD-1/PD-L1 inhibitors despite high neoantigen availability. Consequently, within TLS-high, MSH6-deficient, high-TMB bladder cancers: (i) eIF4E-high tumors will be enriched among non-responders to checkpoint blockade; (ii) eIF4E-low tumors will show comparatively higher response rates. Secondary modifiers (TMB, CD8A, CD3D, PDCD1, CTLA4, TP53, SDC2, TFPI2) will modulate the magnitude of response and the composition of the TLS/immunologic milieu but will not overturn the primary predictor: tumor-intrinsic eIF4E expression within TLS-rich, MSH6-deficient, high-TMB tumors.

| Element | Value |
|---|---|
| Imaging feature | TLS-like lymphoid aggregates |
| Anchor gene(s) | MSH6, eIF4E |
| Primary endpoint | therapy_response |

# 2. Why This Gene — Not Others

## Primary hypothesis gene(s)

**MSH6:** MSH6 (MMR → neoantigen/TMB) axis and the eIF4E (translation-control → immunomodulatory proteome) path because together they provide a mechanistic hypothesis for a mismatch between neoantigen supply and functional immune control.

**eIF4E:** eIF4E (translation-control → immunomodulatory proteome) path because together they provide a mechanistic hypothesis for a mismatch between neoantigen supply and functional immune control.

## Alternative / secondary genes considered

| Tier | Gene | Role / Reason excluded from core |
|---|---|---|
| Secondary | TMB | Supporting modifier from knowledge graph |

| | | |
|---|---|---|
| Secondary | CD8A | Supporting modifier from knowledge graph |
| Secondary | CD3D | Supporting modifier from knowledge graph |
| Secondary | PDCD1 | Supporting modifier from knowledge graph |
| Secondary | CTLA4 | Supporting modifier from knowledge graph |
| Secondary | TP53 | Supporting modifier from knowledge graph |
| Secondary | SDC2 | Supporting modifier from knowledge graph |
| Secondary | TFPI2 | Supporting modifier from knowledge graph |

## 3. Clinically Testable Predictions

**Prediction 1:** At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

**Prediction 2:** In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

## 4. Minimum Viable Validation Plan

### A. Retrospective Cohort Analysis

**Data source:** TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

### B. Tissue-Level Validation

**Methods:** Multiplex IF / IHC / spatial transcriptomics

**Key readouts:** - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

**Goal:** Confirm that the proposed mechanism operates at the spatial / tissue level.

### C. Functional Validation

**Models:** Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

**Perturbations:** Genetic (KO/rescue) or pharmacologic modulation of anchor gene

**Readouts:** T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

**Goal:** Establish causal relationship between anchor gene activity and immune efficacy.

## 5. Strengths, Caveats & Alternative Directions

### Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

### Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

### Alternative Directions

- TMB: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- CD8A: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.